

TECHNICAL REPORT · RENEWABLE ENERGY

Rooftop Solar PV System Design Basis & Cost Estimate

Marriott Hotel – Mezzanine Roof (Drawing D18102-DD-A-104)

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Date	3 November 2026
System capacity	185.4 kWp DC / 160 kW AC, grid-tied, no battery

Document Control

Rev	Date	Description	Prepared	Checked	Approved
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Reference documents

Reference	Title
D18102-DD-A-104	Mezzanine floor plan incl. roof PV zones (architect)
D18102-DD-A-104 Layout1	Mezzanine plan with PV module layout, 10° tilt (this design)
PV_Layout_10deg_Mezzanine_Roof.dxf	PV layout CAD file – modules, tables, setbacks, dimensions

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1. Executive Summary

This report sets out the design basis, layout, energy yield and budget cost for a grid-tied rooftop solar photovoltaic (PV) system on the mezzanine-level roof of the Marriott Hotel, as shown on architectural drawing D18102-DD-A-104. The system has no battery storage and supplies the hotel's own load behind the utility meter.

185.4 kWp	274 MWh	USD 109k	3.4 yrs
DC capacity (299 × 620 Wp)	Year-1 generation	Budget EPC cost (0.59 USD/Wp)	Simple payback

Key findings

- **Capacity.** The two roof zones allocated for PV by the architect (≈1,150 m² gross, including the hatched +4.85 m slab strip) accommodate **299 modules of 620 Wp in 13 rows of 23**, giving **185.4 kWp** at 10° tilt with 800 mm maintenance aisles and 1.0 m edge setbacks.
- **Self-consumption.** The hotel consumes approximately **1,000,000 kWh per month** (average demand ≈1.37 MW). The PV peak output (≈160 kW AC) is around one-tenth of the daytime load, so **100 % of the solar energy is consumed on site**. No energy is exported and no battery is required.
- **Energy and savings.** Year-1 generation is about **274 MWh**, or 2.3 % of the annual consumption. At an avoided tariff of USD 0.12/kWh this saves about **USD 32,923 in year 1**.
- **Cost and return.** The budget turnkey (EPC) cost is **USD 109,278** excluding taxes, including 7 % contingency. Simple payback is **3.4 years**, project IRR 30 % and 25-year NPV (10 %) USD 207,408. The levelised cost of solar energy is about **USD 0.053/kWh**, below the grid tariff.
- **Path to 500 kWp.** The architect's solar zone limits this roof to 185.4 kWp. The balance of ≈315 kWp would need other roof areas or car-park canopies (Section 9).

Recommendation

Proceed with the 185.4 kWp mezzanine roof system as Phase 1, subject to (i) a structural check of both slabs for an added dead load of about 20–25 kg/m² (including ballast) and wind uplift, and (ii) confirmation of the avoided tariff from recent utility bills. In parallel, survey other roof areas and the car park for Phase 2.

2. Site and Design Basis

2.1 Site and load

Facility	Marriott Hotel – twin towers with podium; PV on the mezzanine-level roof
Location	Kampala, Uganda – 0°18'11.34"N, 32°35'13.76"E
Roof zones used	Upper block (+6.80 m SSL, 29.39 × 15.81 m) incl. hatched +4.85 m strip; lower block (architect's PV zone)
Annual electricity consumption	≈12,000,000 kWh (≈1,000,000 kWh/month)
Average / daytime demand	≈1,370 kW / ≈1,700 kW (hotel load profile assumed)
Grid connection	Behind-the-meter, LV main switchboard; zero-export control
Energy storage	None

2.2 Design criteria

Parameter	Value	Basis
Module	620 Wp bifacial TOPCon, 2382 × 1134 × 30 mm, ≈33 kg	Current Tier-1 mainstream size
Orientation / mounting	Portrait, 2 modules high (2P) per table	Matches roof depth
Tilt	10°	Low wind load, self-cleaning by rain
Table depth on plan	4,711 mm (2P); 2,346 mm (1P)	2 × 2382 + 20 mm, × cos 10°
Module clamp gap	20 mm	Mid-clamp
Aisle between tables	800 mm	No inter-row shading 09:00–15:00 all year (≥523 mm required); walkway for cleaning
Edge setback	1,000 mm (east side 1,742 mm)	Wind edge zone, safe access; clear of roof opening
Inverter DC voltage	1,100 V max	Standard C&I string inverter
Design standards	IEC 62548, IEC 60364-7-712, IEC 61215/61730, IEC 62109, IEC 62305	PV array, wiring, modules, inverters, lightning

3. PV Array Layout

The array occupies the two roof zones allocated for PV on drawing D18102-DD-A-104. Each row holds 23 modules (row length 26,522 mm). The upper block carries two 2P tables and one 1P table; the lower block carries four 2P tables.

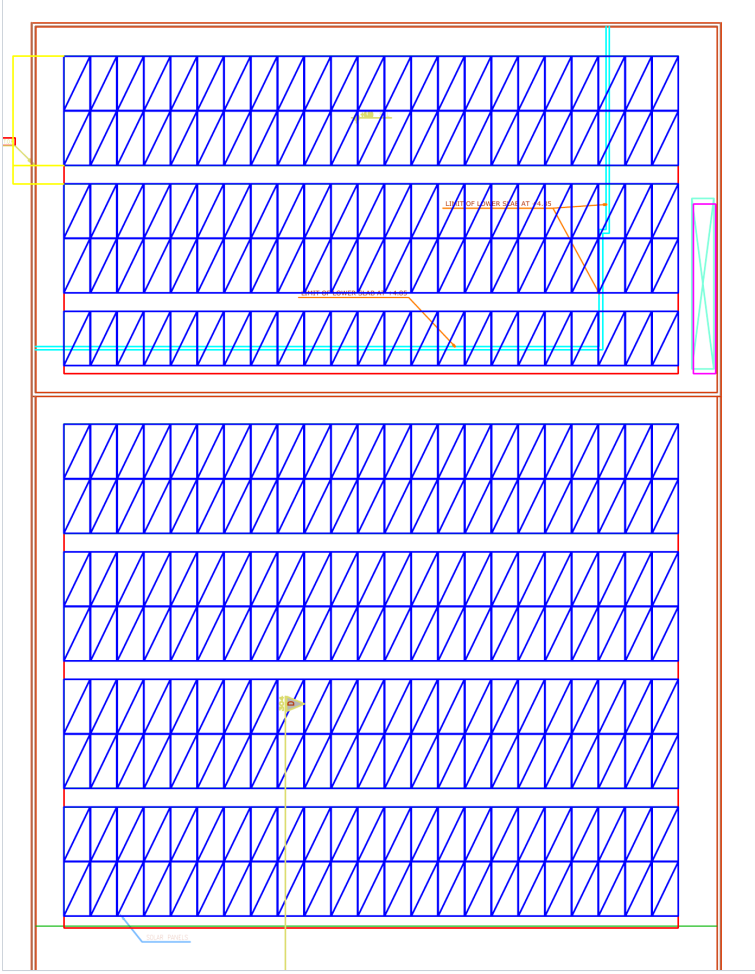


Figure 1 – PV module layout on the mezzanine roof (extract from D18102-DD-A-104 Layout1). Upper block at top, lower block below.

Zone	Tables	Rows	Modules	kWp
Upper block (+6.80 m, incl. hatched +4.85 m strip)	2 × 2P + 1 × 1P	5	115	71.3
Lower block (architect's PV zone)	4 × 2P	8	184	114.1
Total		**13**	**299**	**185.4**

Module area 807 m²; gross roof ≈1,150 m². Indicative added dead load ≈14 kg/m² for modules and structure, plus ballast to suit the wind design.

4. Electrical Design

4.1 Configuration

Item	Inverter INV-01	Inverter INV-02	Total
Rated AC power	80 kW	80 kW	160 kW
Strings	8 × 19 modules	3 × 19 + 5 × 18 modules	16 strings
Modules	152	147	299
DC capacity	94.2 kWp	91.1 kWp	185.4 kWp
DC/AC ratio	1.18	1.14	1.16

4.2 String voltage check

Check	Value	Limit	Result
Max. open-circuit voltage, 19 modules at 12 °C cell temperature (Voc ≈ 50.2 V)	954 V	≤ 1,100 V	OK
Min. MPP voltage, 18 modules at 65 °C cell temperature (Vmp ≈ 36.2 V)	652 V	≥ MPPT minimum (≈200 V)	OK
String current (Isc ≈ 16.1 A, ×1.25 for bifacial gain)	≈20 A	MPPT input rating	OK, max. 2 strings per MPPT

Module electrical values are typical for 620 Wp TOPCon modules. Re-check against the datasheet of the selected module and inverter.

4.3 Single-line arrangement

1. PV strings → 6 mm² double-insulated PV cable in UV-resistant trunking → inverters mounted on the roof in a shaded, ventilated location.
2. Inverters (400 V, 3-phase) → AC combiner panel with outgoing 250 A MCCB, Type II SPD and an isolator at the roof.
3. AC combiner panel → 4C × 120 mm² Cu XLPE/SWA (≈80 m, confirm route) → dedicated 250 A outgoing way on the main LV switchboard.
4. Zero-export controller with CTs on the incomer limits PV output so the site never exports, satisfying utility interface requirements.
5. Earthing: array frames bonded and connected to the building earth; integrated with the existing lightning protection system (IEC 62305).

5. Estimated PV Solar Generation

Site	Kampala, 0°18'11.34"N, 32°35'13.76"E · 185.4 kWp · 10° fixed tilt
Specific yield (year 1, incl. losses)	1,480 kWh/kWp/yr · performance ratio ≈ 0.80
Year-1 generation	**274,362 kWh** · average 22,864 kWh/month · 752 kWh/day
25-year generation	6,477 MWh (1.0 % degradation in year 1, then 0.4 %/yr)
Self-consumption / share of hotel use	100 % · 2.3 % of annual consumption

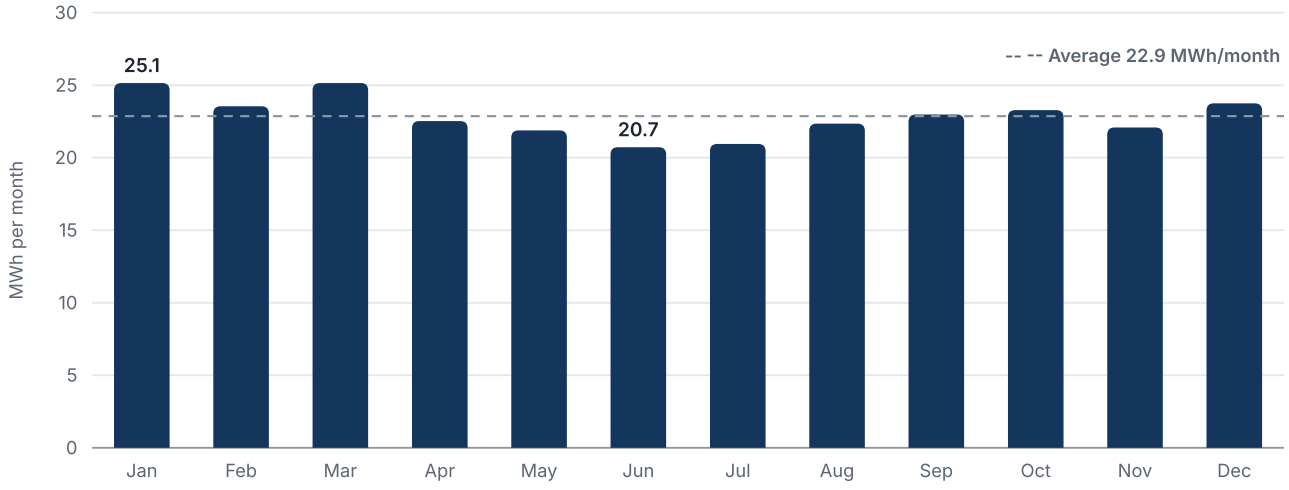


Figure 2 – Estimated monthly PV generation, year 1 (MWh)

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
MWh	25.1	23.6	25.1	22.5	21.9	20.7	21.0	22.3	23.0	23.3	22.1	23.7	**274.4**
kWh/day	811	841	811	751	706	691	676	721	766	751	736	766	752

Output is highest from January to March and lowest in June–July, when cloud cover in Kampala is heaviest. Because the site is on the equator, the difference between the best and worst month is only about 20 %. The hotel load is higher than the PV output in every hour, so all of this energy is used on site.

The monthly split uses typical Kampala solar irradiation; the annual total is the design yield above. Confirm with a PVsyst simulation using the site coordinates at detailed design.

6. Solar Impact on Time-of-Use Tariff

The hotel is billed on a time-of-use (TOU) tariff, so each kWh is worth a different amount depending on when it is used. This section shows when the solar energy is produced and which part of the bill it reduces.

Tariff period	Hours	Indicative rate (USD/kWh)
Off-peak	00:00–06:00	0.07
Shoulder	06:00–18:00	0.12
Peak	18:00–24:00	0.15

Period hours follow the Uganda ERA time-of-use structure; rates are indicative – replace both with the values on the hotel's electricity bill.

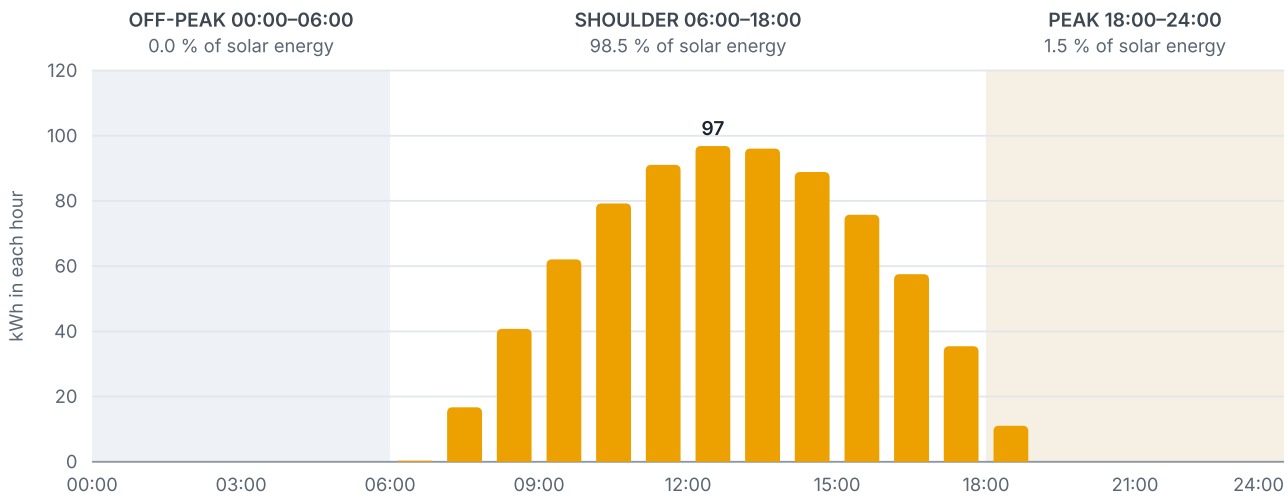


Figure 3 – Average-day solar generation by hour and tariff period (sunrise ≈ 06:50, sunset ≈ 18:56)

Period	Hotel use (MWh/yr)	Solar (MWh/yr)	Share of solar	Grid after solar (MWh/yr)	Reduction	Saving (USD/yr)
Off-peak	2,090	0.0	0.0 %	2,090	0.00 %	0
Shoulder	6,954	270.3	98.5 %	6,684	3.89 %	32,440
Peak	2,956	4.0	1.5 %	2,952	0.14 %	604
Total	**12,000**	**274.4**	**100 %**	**11,726**	**2.29 %**	**33,044**

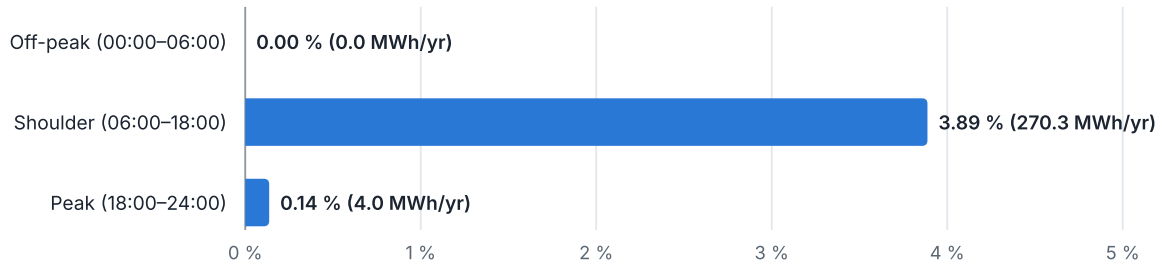


Figure 4 – Reduction of grid energy in each tariff period

What this means

- **Shoulder period:** 98.5 % of the solar energy is produced between 06:00 and 18:00, cutting shoulder-period grid energy by 3.9 %.
- **Peak period:** only 1.5 % falls in the peak period, in the last hour of daylight (18:00–19:00). Solar does not reduce the evening peak, and it does not lower a maximum-demand (kVA) charge set in the evening.
- **Off-peak period:** no solar generation, so no change.
- **Bill impact:** at these rates the energy bill falls by 2.3 %, about USD 33,044 per year. Each solar kWh is worth ≈ USD 0.120, close to the shoulder rate used in the financial analysis.

- Reducing peak-period energy would need battery storage or moving flexible loads (laundry, water heating, pool pumps) into daylight hours.

Average-day model: year-1 solar generation spread over the day at the site's sun times; hotel load assumed at 1,000,000 kWh/month with a typical hotel hourly profile. Confirm with interval-meter data.

7. Cost Estimate (Bill of Quantities)

Budget turnkey (EPC) estimate in US dollars, based on 2026 East African market prices for commercial rooftop systems. Accuracy class ±15 %.

No.	Description	Qty	Rate	Amount (USD)
1	PV modules, 620 Wp bifacial TOPCon, Tier-1 (landed, cleared)	299 nos	USD 0.150/Wp	27,807
2	String inverters, 80 kW, 1,100 V, multi-MPPT (incl. comms)	2 nos	USD 4,800 each	9,600
3	Mounting structure, aluminium, 10° tilt, ballasted/anchored, incl. roof pads	185.4 kWp	USD 0.090/Wp	16,684
4	DC system: 6 mm² PV cable, MC4 connectors, DC isolators, SPD Type II	Lot	USD 0.040/Wp	7,415
5	AC system: AC combiner panel, 250 A MCCB, 4C×120 mm² Cu XLPE to main LV board (≈80 m), cable tray	Lot	Lump sum	9,500
6	Zero-export controller, CT/power meter, data logger & remote monitoring	1 set	Lump sum	3,500
7	Earthing, equipotential bonding & lightning protection integration	Lot	Lump sum	3,000
8	Roof walkways, safety lifeline & water point for cleaning	Lot	Lump sum	3,500
9	Installation, testing & commissioning (labour, supervision, lifting)	185.4 kWp	USD 0.060/Wp	11,123
10	Engineering: detailed design, structural check of slab, PVsyst, as-builts	Lot	Lump sum	4,000
11	Approvals & utility interface (ERA / UEDCL), O&M manuals, training	Lot	Lump sum	2,000
12	Transport, logistics & customs clearing	Lot	Lump sum	4,000
	Sub-total			**102,129**
	Contingency (7 %)			7,149
	Total budget EPC cost (excluding taxes)			**109,278**
	Specific cost			0.589 USD/Wp

Excludes VAT, withholding tax and import duties (solar equipment is generally zero-rated or exempt in Uganda; confirm with URA), roof waterproofing upgrades, structural strengthening if required, and any utility charges. Annual O&M (cleaning, inspection, monitoring) is estimated at USD 1,311/yr (≈1.2 % of capex), escalating at 3 %/yr.

8. Financial Analysis

Investment (EPC, excluding taxes)	USD 109,278
Avoided grid tariff (year 1) / escalation	USD 0.12/kWh / 2 % per year
Year-1 savings	USD 32,923
Inverter replacement (year 12)	USD 9,600
Simple payback	3.4 years
Project IRR (25 years, unlevered)	30.1 %
NPV at 10 % discount rate	USD 207,408
Levelised cost of energy (LCOE)	USD 0.053/kWh
25-year cumulative savings, net of O&M	USD 934,239

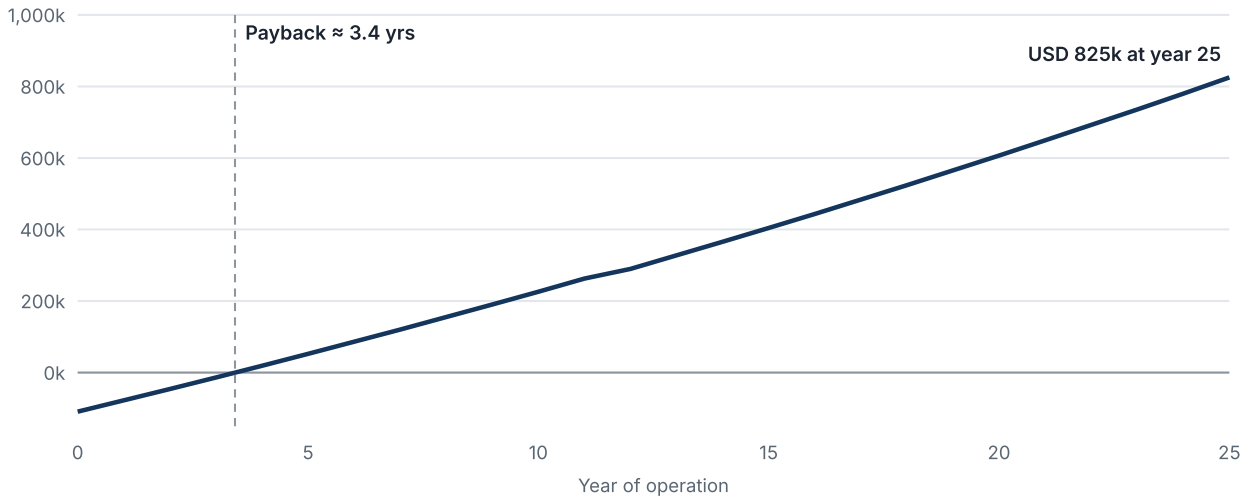


Figure 5 – Cumulative net cash flow (USD)

Tariff sensitivity

Avoided tariff (USD/kWh)	Payback (yrs)	IRR	NPV @10 % (USD)
0.09	4.6	22%	123,694
0.10	4.1	25%	151,599
0.12	**3.4**	**30%**	**207,408**
0.14	2.9	35%	263,217
0.16	2.5	40%	319,026

Annual cash flow (selected years, USD)

Year	Generation (MWh)	Tariff (USD/kWh)	Savings	O&M	Replacement	Net cash flow	Cumulative
0						-109,278	-109,278
1	274.4	0.120	32,923	1,311	–	31,612	-77,666
2	270.5	0.122	33,113	1,350	–	31,763	-45,903
3	269.4	0.125	33,640	1,391	–	32,249	-13,653
4	268.4	0.127	34,176	1,433	–	32,743	19,090
5	267.3	0.130	34,720	1,476	–	33,244	52,334
6	266.2	0.133	35,273	1,520	–	33,753	86,087
8	264.1	0.138	36,405	1,612	–	34,792	155,149
10	262.0	0.143	37,573	1,711	–	35,863	226,335
12	259.9	0.149	38,779	1,815	9,600	27,364	290,108
15	256.8	0.158	40,661	1,983	–	38,678	404,411
20	251.7	0.175	44,002	2,299	–	41,703	606,789
25	246.7	0.193	47,618	2,665	–	44,953	824,961

9. Expansion Options

The target of about 500 kWp exceeds what the mezzanine roof can hold. Because the hotel load is around ten times the PV output, every additional kWp would also be fully self-consumed. Expansion can be added in stages without a battery.

Option	Capacity	Year-1 energy	Budget cost (USD)	Remarks
Phase 1 – mezzanine roof (this report)	185.4 kWp	274 MWh	109,278	Ready for detailed design
Other roofs (left podium roof ≈370 m², lobby roof)	≈ 65–120 kWp	≈ 95–180 MWh	≈ 38,000–68,000	Survey and structural check required
Car-park solar canopies	≈ 150–250 kWp	≈ 220–370 MWh	≈ 0.75–0.85 USD/Wp	Adds shaded parking; steel canopy cost
Indicative total	**≈ 400–555 kWp**	**≈ 590–825 MWh**		**≈ 5–7 % of hotel consumption**

10. Assumptions, Exclusions and Next Steps

10.1 Key assumptions

- Roof geometry is taken from drawing D18102-DD-A-104; the hatched +4.85 m slab strip is available for PV, as confirmed by the client.
- Both roof slabs can carry the PV dead load and wind loads without strengthening, subject to verification by the structural engineer.
- The hotel load is about 1,000,000 kWh per month with a continuous daytime demand well above 160 kW.
- The avoided tariff is USD 0.12/kWh escalating at 2 %/yr, to be confirmed from the latest utility bills (energy plus demand components).
- Prices are 2026 East African market levels in USD, excluding taxes. The exchange rate is not considered.

10.2 Exclusions

- Battery storage, diesel-generator synchronisation and export metering.
- Roof waterproofing renewal, structural strengthening, and civil works to the LV room.
- Utility network upgrades, permit fees and financing costs.

10.3 Recommended next steps

1. Obtain 12 months of utility bills or interval-meter data and confirm the tariff structure.
2. Commission a structural assessment of both slabs (dead load, ballast, wind uplift) and agree the roof fixing method with the architect.
3. Confirm the inverter location, AC cable route and spare way on the main LV switchboard.
4. Run a PVsyst simulation with site coordinates and the selected module and inverter datasheets.
5. Issue the tender package (layout, single-line diagram, specification, BOQ) to at least three qualified EPC contractors.
6. Start the Phase 2 survey of other roofs and the car park to move towards the 500 kWp target.